Sequences, Series & Complex Numbers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic sequences**

Find the 12th term of the arithmetic sequence $5,\ 9,\ 13,\ \ldots$

$a_1 = 5$, $d = 4$. $a_{12} = 5 + (11)(4) = 49$.

2 **[Foundational]** • **Target: Arithmetic sequences**

The 4th term of an arithmetic sequence is 18, and the 9th term is 38. Find the first term and the common difference.

$a_1 + 3d = 18$ and $a_1 + 8d = 38$. Subtracting: $5d = 20 \Rightarrow d = 4$, so $a_1 = 18 - 12 = 6$.

3 **[Foundational]** • **Target: Arithmetic sequences**

Find the sum of the first 20 terms of the arithmetic sequence $7,\ 11,\ 15,\ \ldots$

$a_1 = 7$, $d = 4$, $a_{20} = 7 + (19)(4) = 83$. $S_{20} = \frac{20}{2}(7 + 83) = 10(90) = 900$.

4 **[Foundational]** • **Target: Arithmetic sequences**

Find the number of terms in the arithmetic sequence $6,\ 10,\ 14,\ \ldots,\ 150$.

$a_1 = 6$, $d = 4$. $6 + 4(n - 1) = 150 \Rightarrow 4(n - 1) = 144 \Rightarrow n - 1 = 36 \Rightarrow n = 37$.

5 **[Foundational]** • **Target: Geometric sequences**

Find the 7th term of the geometric sequence $3,\ 6,\ 12,\ \ldots$

$a_1 = 3$, $r = 2$. $a_7 = 3(2^6) = 3(64) = 192$.

6 **[Foundational]** • **Target: Geometric sequences**

In a geometric sequence with all positive terms, the 3rd term is 20 and the 5th term is 80. Find r and a_1.

$a_1 r^2 = 20$ and $a_1 r^4 = 80$. Dividing: $r^2 = 4$, and since all terms are positive, $r = 2$. Then $a_1 = \frac{20}{4} = 5$.

7 **[Foundational]** • **Target: Geometric sequences**

Given that $x - 1$, $x + 1$, and $3x - 1$ are consecutive terms of a geometric sequence, find the possible value(s) of x.

$(x + 1)^2 = (x - 1)(3x - 1) \Rightarrow x^2 + 2x + 1 = 3x^2 - 4x + 1 \Rightarrow 0 = 2x^2 - 6x = 2x(x - 3)$. So $x = 0$ or $x = 3$. If $x = 0$, the terms are $-1,\ 1,\ -1$ (ratio -1, valid). If $x = 3$, the terms are $2,\ 4,\ 8$ (ratio 2, valid).

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric sequences

Find the sum of the first 6 terms of the geometric sequence $5,\ 15,\ 45,\ \ldots$

$a_1 = 5, r = 3$. $S_6 = \frac{5(3^6 - 1)}{3 - 1} = \frac{5(728)}{2} = 1820$.

9 [Intermediate] • Target: Series and sigma notation

Evaluate $\sum_{k=1}^{5} (3k - 2)$.

Terms: $1, 4, 7, 10, 13$. Sum $= 1 + 4 + 7 + 10 + 13 = 35$.

10 [Intermediate] • Target: Series and sigma notation

Write the series $3 + 7 + 11 + \cdots + 39$ in sigma notation, then evaluate it.

The general term is $a_k = 4k - 1$ for $k = 1, \ldots, 10$ (since $4(10) - 1 = 39$). $\sum_{k=1}^{10} (4k - 1) = 210$.

11 [Intermediate] • Target: Series and sigma notation

Evaluate $\sum_{n=1}^{6} 2^n$.

$2 + 4 + 8 + 16 + 32 + 64 = 126$.

12 [Intermediate] • Target: Geometric series

Find the sum of the first 8 terms of the geometric sequence $2,\ 6,\ 18,\ \ldots$

$a_1 = 2, r = 3$. $S_8 = \frac{2(3^8 - 1)}{3 - 1} = \frac{2(6561 - 1)}{2} = 6560$.

13 [Intermediate] • Target: Geometric series

Find the sum to infinity of $16 - 8 + 4 - 2 + \cdots$

$a_1 = 16, r = -\frac{1}{2}$. $S_\infty = \frac{16}{1 - \left(-\frac{1}{2}\right)} = \frac{16}{3/2} = \frac{32}{3}$.

14 [Intermediate] • Target: Geometric series

Express the repeating decimal $0.4\overline{5}$ (that is, $0.45555\ldots$) as a geometric series and find its exact fractional value.

$0.4\overline{5} = 0.4 + (0.05 + 0.005 + 0.0005 + \cdots)$. The bracketed part is geometric with $a_1 = 0.05$, $r = 0.1$, summing to $\frac{0.05}{0.9} = \frac{1}{18}$. Total: $\frac{2}{5} + \frac{1}{18} = \frac{36}{90} + \frac{5}{90} = \frac{41}{90}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Given $z_1 = 5 - 2i$ and $z_2 = -3 + 4i$, find $z_1 + z_2$ and $z_1 - z_2$.

$z_1 + z_2 = (5 - 3) + (-2 + 4)i = 2 + 2i$. $z_1 - z_2 = (5 - (-3)) + (-2 - 4)i = 8 - 6i$.

16 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 5.01 Trigonometry

Multiply $(3 + 2i)(4 - 5i)$.

$12 - 15i + 8i - 10i^2 = 12 - 7i + 10 = 22 - 7i$.

17 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Simplify i^{23}, using $i^2 = -1$.

Powers of i cycle with period 4: $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. Since $23 = 4(5) + 3$, $i^{23} = i^3 = -i$.

18 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 7.01 Solid Geometry & Similarity

Simplify $(2 - 3i) - (5 + i) + (1 - 4i)$.

$(2 - 5 + 1) + (-3 - 1 - 4)i = -2 - 8i$.

19 [Challenging] • Target: Complex conjugates and division • Cross-Lesson: 8.01 Statistics & Probability

Find the conjugate of $z = 6 + 7i$, and compute $z\bar{z}$.

$\bar{z} = 6 - 7i$. $z\bar{z} = 6^2 + 7^2 = 36 + 49 = 85$.

20 [Challenging] • Target: Complex conjugates and division • Cross-Lesson: 1.01 Advanced Algebra

Simplify $\frac{4 + 3i}{1 - 2i}$ by multiplying by the conjugate of the denominator.

$\frac{4 + 3i}{1 - 2i} \cdot \frac{1 + 2i}{1 + 2i} = \frac{(4 + 3i)(1 + 2i)}{1^2 + 2^2} = \frac{4 + 8i + 3i + 6i^2}{5} = \frac{-2 + 11i}{5} = -\frac{2}{5} + \frac{11}{5}i$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic sequences

Find the 10th term of the arithmetic sequence $10, 16, 22, \ldots$

a. 66

b. 70

c. 64 ✓

d. 58

$a_1 = 10$, $d = 6$. $a_{10} = 10 + (9)(6) = 64$.

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22 [Foundational] • Target: Geometric sequences

Find the 14th term of the arithmetic sequence $3,\ 5,\ 7,\ \ldots$

a. 29 ✓

b. 31

c. 27

d. 32

$a_1 = 3, d = 2.\ a_{14} = 3 + (13)(2) = 29.$

23 [Foundational] • Target: Series and sigma notation

Find the 11th term of the arithmetic sequence $-3,\ -6,\ -9,\ \ldots$

a. -36

b. -30

c. -31

d. -33 ✓

$a_1 = -3, d = -3.\ a_{11} = -3 + (10)(-3) = -33.$

24 [Foundational] • Target: Geometric series

Find the 13th term of the arithmetic sequence $-13,\ -7,\ -1,\ \ldots$

a. 53

b. 65

c. 61

d. 59 ✓

$a_1 = -13, d = 6.\ a_{13} = -13 + (12)(6) = 59.$

25 [Foundational] • Target: Complex number arithmetic

Find the 15th term of the arithmetic sequence $12,\ 8,\ 4,\ \ldots$

a. -40

b. -48

c. -44 ✓

d. -42

$a_1 = 12, d = -4.\ a_{15} = 12 + (14)(-4) = -44.$

26 [Foundational] • Target: Complex conjugates and division

Find the 10th term of the arithmetic sequence $11,\ 17,\ 23,\ \ldots$

a. 59

b. 67

c. 65 ✓

d. 71

$a_1 = 11, d = 6.\ a_{10} = 11 + (9)(6) = 65.$

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27 [Foundational] • Target: Modulus of complex numbers

Find the 10th term of the arithmetic sequence $11,\ 14,\ 17,\ \ldots$

a. 35

b. 38 ✓

c. 40

d. 41

$a_1 = 11, d = 3.\ a_{10} = 11 + (9)(3) = 38.$

28 [Foundational] • Target: Quadratic equations with complex roots

Find the 4th term of the geometric sequence $-1,\ 2,\ -4,\ \ldots$

a. 6

b. 8 ✓

c. 10

d. -8

$a_1 = -1, r = -2.\ a_4 = -1((-2)^3) = -1(-8) = 8.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a bouncing ball

Find the 6th term of the geometric sequence $-6,\ -12,\ -24,\ \ldots$

a. -192 ✓

b. -190

c. 192

d. -194

$a_1 = -6, r = 2.\ a_6 = -6(2^5) = -6(32) = -192.$

30 [Intermediate] • Target: Arithmetic & Geometric Progressions

Find the 7th term of the geometric sequence $-8,\ -24,\ -72,\ \ldots$

a. -5830

b. 5832

c. -5834

d. -5832 ✓

$a_1 = -8, r = 3.\ a_7 = -8(3^6) = -8(729) = -5832.$

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31 [Intermediate] • Target: Binomial Expansions & General Terms

Find the 3rd term of the geometric sequence $7,\ 21,\ 63,\ \ldots$

a. 63 ✓

b. 61

c. -63

d. 65

$a_1 = 7, r = 3.\ a_3 = 7(3^2) = 7(9) = 63.$

32 [Intermediate] • Target: Operations in the Complex Plane

Find the 6th term of the geometric sequence $8,\ -24,\ 72,\ \ldots$

a. -1946

b. -1944 ✓

c. -1942

d. 1944

$a_1 = 8, r = -3.\ a_6 = 8((-3)^5) = 8(-243) = -1944.$

33 [Intermediate] • Target: De Moivre's Formula & Roots

Find the 3rd term of the geometric sequence $-6,\ 12,\ -24,\ \ldots$

a. 24

b. -26

c. -24 ✓

d. -22

$a_1 = -6, r = -2.\ a_3 = -6((-2)^2) = -6(4) = -24.$

34 [Intermediate] • Target: Arithmetic sequences

Find the 4th term of the geometric sequence $-1,\ -2,\ -4,\ \ldots$

a. -10

b. 8

c. -8 ✓

d. -6

$a_1 = -1, r = 2.\ a_4 = -1(2^3) = -1(8) = -8.$

35 [Intermediate] • Target: Geometric sequences

Find the sum of the first 4 terms of the geometric sequence $-9,\ -18,\ -36,\ \ldots$

a. -135 ✓

b. -133

c. -144

d. -137

$a_1 = -9, r = 2.\ S_4 = \frac{-9(2^4 - 1)}{2 - 1} = \frac{-9(15)}{1} = -135.$

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36 [Intermediate] • Target: Series and sigma notation

Find the sum of the first 7 terms of the geometric sequence $-4,\ 12,\ -36, \ldots$

a. -2188 ✓

b. -2192

c. -2190

d. -2186

$a_1 = -4, r = -3. S_7 = \frac{-4((-3)^7 - 1)}{-3 - 1} = \frac{-4(-2188)}{-4} = -2188.$

37 [Intermediate] • Target: Geometric series

Multiply $(2 - 7i)(5 + 3i)$.

a. $31 - 27i$

b. $31 - 29i$ ✓

c. $31 + 41i$

d. $-11 - 29i$

$10 + 6i - 35i - 21i^2 = 10 - 29i - 21(-1) = 31 - 29i.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 7.01 Solid Geometry & Similarity

Multiply $(5 + 5i)(-6 - 4i)$.

a. $-50 - 50i$

b. $-10 - 48i$

c. $-10 - 50i$ ✓

d. $-10 + 10i$

$-30 - 20i - 30i - 20i^2 = -30 - 50i - 20(-1) = -10 - 50i.$

39 [Challenging] • Target: Complex conjugates and division • Cross-Lesson: 8.01 Statistics & Probability

Find $|z|$ for $z = 6 - 8i$.

a. 8

b. 10 ✓

c. 14

d. 12

$|z| = \sqrt{(6)^2 + (-8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$

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40 [Challenging] • Target: Modulus of complex numbers • Cross-Lesson: 1.01 Advanced Algebra

Find $|z|$ for $z = -11 + 60i$.

a. 59

b. 61 ✓

c. 71

d. 63

$|z| = \sqrt{(-11)^2 + (60)^2} = \sqrt{121 + 3600} = \sqrt{3721} = 61.$

41 [Challenging] • Target: Arithmetic & Geometric Progressions • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In an arithmetic progression, the sum of the first n terms is given by $S_n = 3n^2 + 2n$. What is the 10th term a_{10}?

a. 59 ✓

b. 62

c. 65

d. 320

$a_{10} = S_{10} - S_9 = [3(100) + 20] - [3(81) + 18] = 320 - [243 + 18] = 320 - 261 = 59.$

42 [Challenging] • Target: Arithmetic & Geometric Progressions • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Evaluate the sum of the convergent geometric series: $\sum_{n=0}^{\infty} 5(-\frac{1}{3})^n$.

a. $\frac{15}{4}$ ✓

b. $\frac{15}{2}$

c. $\frac{5}{4}$

d. 5

First term $a = 5(-1/3)^0 = 5$, common ratio $r = -\frac{1}{3}$. Sum $= \frac{a}{1-r} = \frac{5}{1-(-1/3)} = \frac{5}{4/3} = \frac{15}{4}$.

43 [Challenging] • Target: Binomial Expansions & General Terms • Cross-Lesson: 5.01 Trigonometry

What is the sum of the coefficients in the expansion of $(5x - 4)^7$?

a. 1 ✓

b. 128

c. 2187

d. 0

Substitute $x = 1$: $(5(1) - 4)^7 = 1^7 = 1$.

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44 [Challenging] • Target: Operations in the Complex Plane • Cross-Lesson: 6.01 Circles & Coordinate Geometry

What is the modulus of the complex number $z = (3 + 4i)(5 - 12i)$?

a. 65 ✓

b. 17

c. 13

d. 25

$|z| = |3 + 4i| \times |5 - 12i| = \sqrt{3^2 + 4^2} \times \sqrt{5^2 + (-12)^2} = 5 \times 13 = 65$.

45 [Challenging] • Target: De Moivre's Formula & Roots • Cross-Lesson: 7.01 Solid Geometry & Similarity

Evaluate $(\cos 30^\circ + i\sin 30^\circ)^6$.

a. -1 ✓

b. 1

c. i

d. $-i$

By De Moivre's: $\cos(6 \times 30^\circ) + i\sin(6 \times 30^\circ) = \cos 180^\circ + i\sin 180^\circ = -1 + 0i = -1$.

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